2.5 Entity-Relationship Diagram

As we saw briefly in Section 1.4, an **E-R diagram** can express the overall logical structure of a database graphically. E-R diagrams are simple and clear—qualities that may well account in large part for the widespread use of the E-R model. Such a diagram consists of the following major components:

- **Rectangles**, which represent entity sets
- **Ellipses**, which represent attributes
- **Diamonds**, which represent relationship sets
- **Lines**, which link attributes to entity sets and entity sets to relationship sets
- **Double ellipses**, which represent multivalued attributes
- **Dashed ellipses**, which denote derived attributes
- **Double lines**, which indicate total participation of an entity in a relationship set
- **Double rectangles**, which represent weak entity sets (described later, in Section 2.6.)

Consider the entity-relationship diagram in Figure 2.8, which consists of two entity sets, *customer* and *loan*, related through a binary relationship set *borrower*. The attributes associated with *customer* are *customer-id*, *customer-name*, *customer-street*, and *customer-city*. The attributes associated with *loan* are *loan-number* and *amount*. In Figure 2.8, attributes of an entity set that are members of the primary key are underlined.

The relationship set *borrower* may be many-to-many, one-to-many, many-to-one, or one-to-one. To distinguish among these types, we draw either a directed line (→) or an undirected line (—) between the relationship set and the entity set in question.

- A directed line from the relationship set *borrower* to the entity set *loan* specifies that *borrower* is either a one-to-one or many-to-one relationship set, from *customer* to *loan*; *borrower* cannot be a many-to-many or a one-to-many relationship set from *customer* to *loan*.

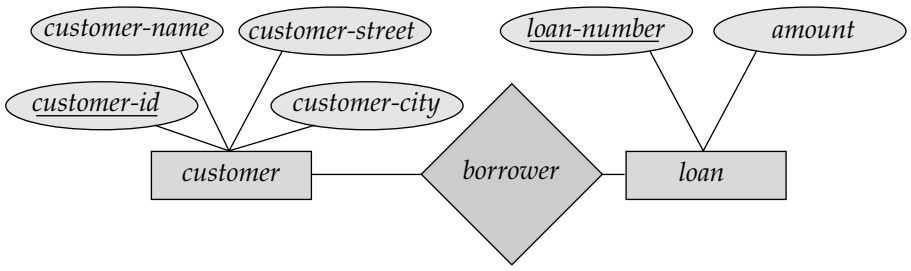


Figure 2.8 E-R diagram corresponding to customers and loans.

- An undirected line from the relationship set *borrower* to the entity set *loan* specifies that *borrower* is either a many-to-many or one-to-many relationship set from *customer* to *loan*.

Returning to the E-R diagram of Figure 2.8, we see that the relationship set *borrower* is many-to-many. If the relationship set *borrower* were one-to-many, from *customer* to *loan*, then the line from *borrower* to *customer* would be directed, with an arrow pointing to the *customer* entity set (Figure 2.9a). Similarly, if the relationship set *borrower* were many-to-one from *customer* to *loan*, then the line from *borrower* to *loan* would have an arrow pointing to the *loan* entity set (Figure 2.9b). Finally, if the relationship set *borrower* were one-to-one, then both lines from *borrower* would have arrows:

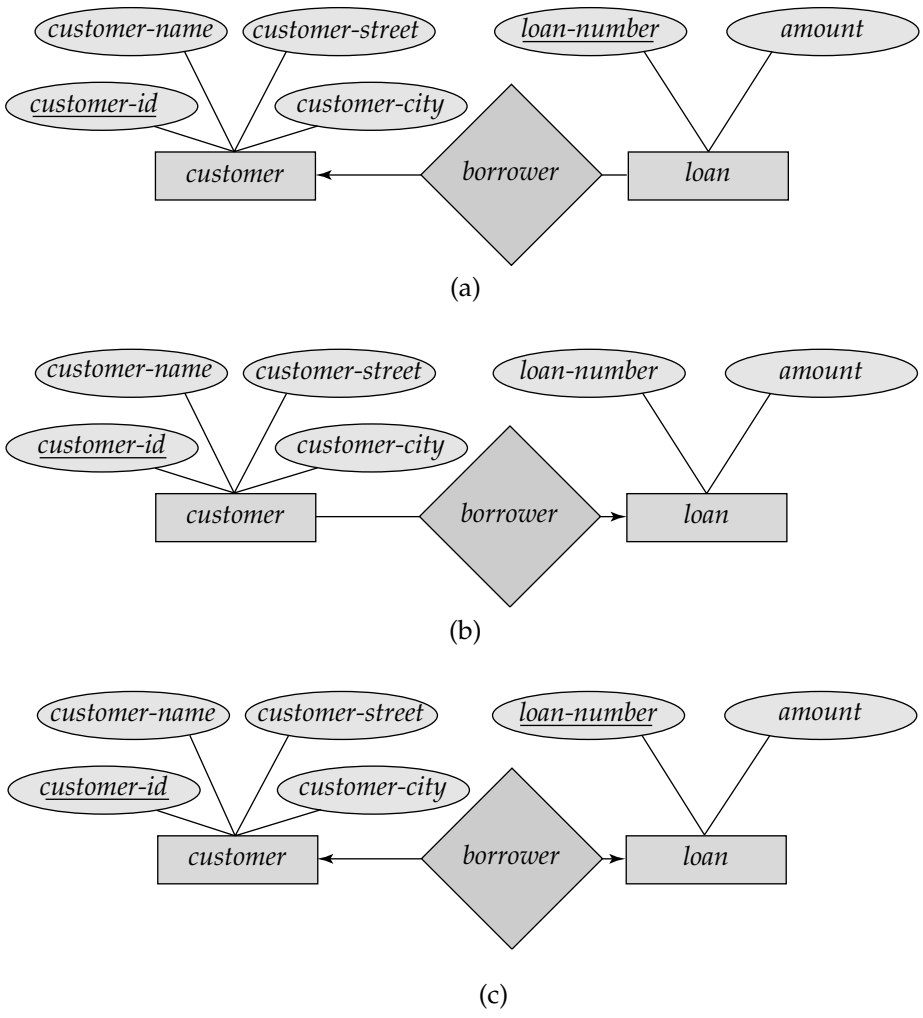


Figure 2.9 Relationships. (a) one to many. (b) many to one. (c) one-to-one.

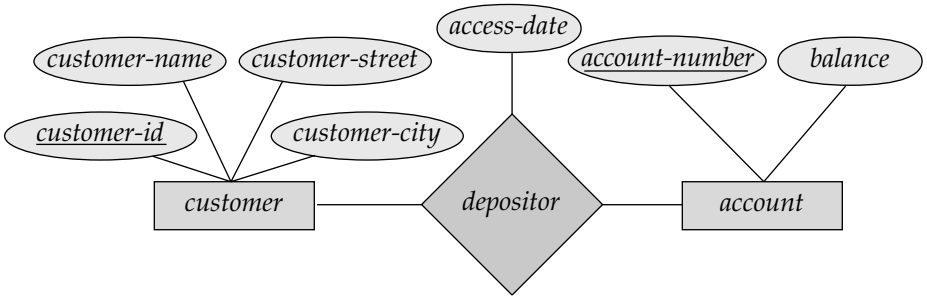


Figure 2.10 E-R diagram with an attribute attached to a relationship set.

one pointing to the *loan* entity set and one pointing to the *customer* entity set (Figure 2.9c).

If a relationship set has also some attributes associated with it, then we link these attributes to that relationship set. For example, in Figure 2.10, we have the *access-date* descriptive attribute attached to the relationship set *depositor* to specify the most recent date on which a customer accessed that account.

Figure 2.11 shows how composite attributes can be represented in the E-R notation. Here, a composite attribute *name*, with component attributes *first-name*, *middle-initial*, and *last-name* replaces the simple attribute *customer-name* of *customer*. Also, a composite attribute *address*, whose component attributes are *street*, *city*, *state*, and *zip-code* replaces the attributes *customer-street* and *customer-city* of *customer*. The attribute *street* is itself a composite attribute whose component attributes are *street-number*, *street-name*, and *apartment number*.

Figure 2.11 also illustrates a multivalued attribute *phone-number*, depicted by a double ellipse, and a derived attribute *age*, depicted by a dashed ellipse.

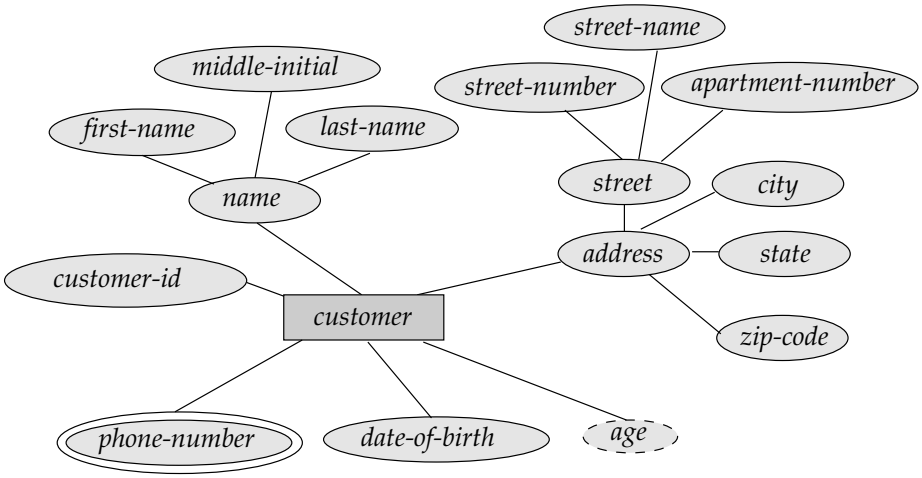


Figure 2.11 E-R diagram with composite, multivalued, and derived attributes.

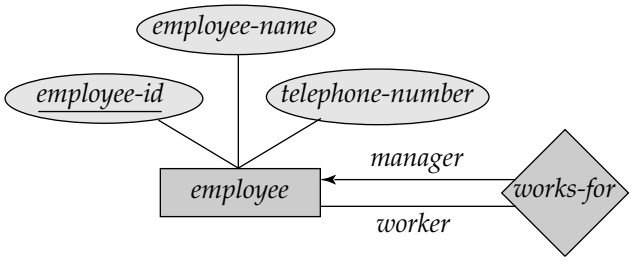


Figure 2.12 E-R diagram with role indicators.

We indicate roles in E-R diagrams by labeling the lines that connect diamonds to rectangles. Figure 2.12 shows the role indicators *manager* and *worker* between the *employee* entity set and the *works-for* relationship set.

Nonbinary relationship sets can be specified easily in an E-R diagram. Figure 2.13 consists of the three entity sets *employee*, *job*, and *branch*, related through the relationship set *works-on*.

We can specify some types of many-to-one relationships in the case of nonbinary relationship sets. Suppose an employee can have at most one job in each branch (for example, Jones cannot be a manager and an auditor at the same branch). This constraint can be specified by an arrow pointing to *job* on the edge from *works-on*.

We permit at most one arrow out of a relationship set, since an E-R diagram with two or more arrows out of a nonbinary relationship set can be interpreted in two ways. Suppose there is a relationship set R between entity sets $A_1, A_2, \ldots, A_n$, and the only arrows are on the edges to entity sets $A_{i+1}, A_{i+2}, \ldots, A_n$. Then, the two possible interpretations are:

1. A particular combination of entities from $A_1, A_2, \ldots, A_i$ can be associated with at most one combination of entities from $A_{i+1}, A_{i+2}, \ldots, A_n$. Thus, the primary key for the relationship R can be constructed by the union of the primary keys of $A_1, A_2, \ldots, A_i$.

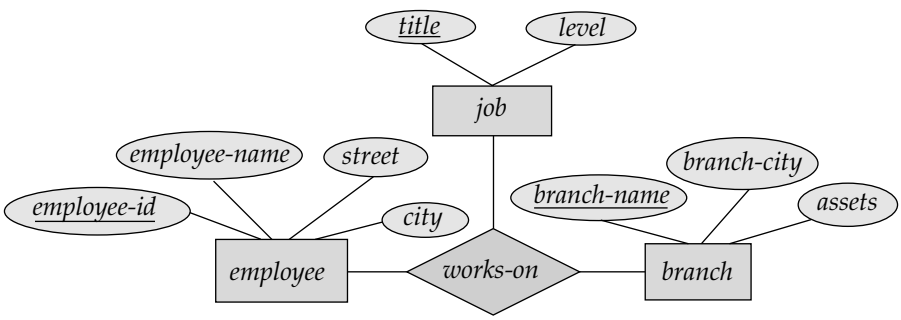


Figure 2.13 E-R diagram with a ternary relationship.

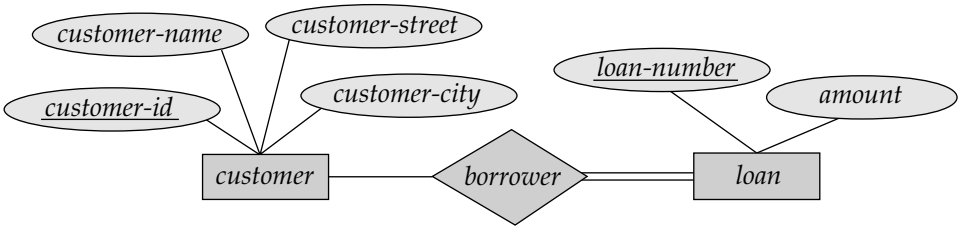


Figure 2.14 Total participation of an entity set in a relationship set.

2. For each entity set A_k, $i < k \leq n$, each combination of the entities from the other entity sets can be associated with at most one entity from A_k. Each set $\{A_1, A_2, \ldots, A_{k-1}, A_{k+1}, \ldots, A_n\}$, for $i < k \leq n$, then forms a candidate key.

Each of these interpretations has been used in different books and systems. To avoid confusion, we permit only one arrow out of a relationship set, in which case the two interpretations are equivalent. In Chapter 7 (Section 7.3) we study the notion of *functional dependencies*, which allow either of these interpretations to be specified in an unambiguous manner.

Double lines are used in an E-R diagram to indicate that the participation of an entity set in a relationship set is total; that is, each entity in the entity set occurs in at least one relationship in that relationship set. For instance, consider the relationship *borrower* between customers and loans. A double line from *loan* to *borrower*, as in Figure 2.14, indicates that each loan must have at least one associated customer.

E-R diagrams also provide a way to indicate more complex constraints on the number of times each entity participates in relationships in a relationship set. An edge between an entity set and a binary relationship set can have an associated minimum and maximum cardinality, shown in the form $l..h$, where l is the minimum and h the maximum cardinality. A minimum value of 1 indicates total participation of the entity set in the relationship set. A maximum value of 1 indicates that the entity participates in at most one relationship, while a maximum value $*$ indicates no limit. Note that a label $1..*$ on an edge is equivalent to a double line.

For example, consider Figure 2.15. The edge between *loan* and *borrower* has a cardinality constraint of $1..1$, meaning the minimum and the maximum cardinality are both 1. That is, each loan must have exactly one associated customer. The limit $0..*$ on the edge from *customer* to *borrower* indicates that a customer can have zero or more loans. Thus, the relationship *borrower* is one to many from *customer* to *loan*, and further the participation of *loan* in *borrower* is total.

It is easy to misinterpret the $0..*$ on the edge between *customer* and *borrower*, and think that the relationship *borrower* is many to one from *customer* to *loan*—this is exactly the reverse of the correct interpretation.

If both edges from a binary relationship have a maximum value of 1, the relationship is one to one. If we had specified a cardinality limit of $1..*$ on the edge between *customer* and *borrower*, we would be saying that each customer must have at least one loan.